

Real Analysis: 1.3

Theresa Wunderlich

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1 1.5 Order and Absolute Value

Definition 1.5. We define an order relation on $\mathbb{R}$ as follows: for all real numbers a and b we say that a is *less than* b ($a < b$) if $b - a \in \mathbb{R}^+$. In other words, $a < b$ means that the number $b - a$ is positive. For all real numbers a and b , we say that $a \leq b$ if $b - a \in \mathbb{R}^+ \cup \{0\}$. In other words, $a \leq b$ means that the number $b - a$ is positive or zero. The symbols $>$ and $\geq$ are defined similarly.

Problem 1. Prove that for all real numbers a , b , and c :

- (a) $a \leq a$.
- (b) If $a \leq b$ and $b \leq a$, then $a = b$.
- (c) If $a \leq b$ and $b \leq c$, then $a \leq c$.

Proof of (a). By definition, $a \leq a$ means $a - a \in \mathbb{R}^+ \cup \{0\}$. Since $a - a = 0 \in \mathbb{R}^+ \cup \{0\}$, we conclude that $a \leq a$. $\square$

Proof of (b). By definition, $a \leq b$ means $b - a \in \mathbb{R}^+ \cup \{0\}$, and $b \leq a$ means $a - b \in \mathbb{R}^+ \cup \{0\}$. We split into two cases according to the value of $b - a$.

Case 1: $b - a = 0$. Adding a to both sides gives $b = a$, so $a = b$.

Case 2: $b - a \in \mathbb{R}^+$. Since $a - b = -(b - a)$, the number $a - b$ is negative, i.e. $a - b \in \mathbb{R}^-$. However, $b \leq a$ gives $a - b \in \mathbb{R}^+ \cup \{0\}$, so $a - b$ is positive or zero. No number is both negative and nonnegative by our previous proof of Axiom 1.4, so this is a contradiction; therefore Case 2 cannot occur.

So, we must have $b - a = 0$, and therefore $a = b$. $\square$

Proof of (c). By definition, $a \leq b$ means $b - a \in \mathbb{R}^+ \cup \{0\}$, and $b \leq c$ means $c - b \in \mathbb{R}^+ \cup \{0\}$. Adding these two quantities,

$$(b - a) + (c - b) = c - a.$$

The sum of two elements of $\mathbb{R}^+ \cup \{0\}$ again lies in $\mathbb{R}^+ \cup \{0\}$ (the sum of two positive numbers is positive, and adding 0 changes nothing). Hence $c - a \in \mathbb{R}^+ \cup \{0\}$, which means $a \leq c$. $\square$